VECTRA: The Quarantine Matrix

Constraining Neural Hallucinations in 3D Gaussian Environments

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1. Introduction

- **The Spatial AI Convergence**

I will outline how generative artificial intelligence and high-fidelity spatial computing are rapidly converging. I will then introduce the Vectra Protocol as a localized, secure architecture designed to seamlessly integrate generated 3D assets into real-world digital twins.

2. Motivation

- **Overcoming Hardware and Spatial Bottlenecks**

Traditional 3D generative integration suffers from severe computational bottlenecks and destructive spatial overlapping, often leading to out-of-memory kernel panics on edge hardware. I aim to demonstrate how engineering a memory-efficient, non-destructive volumetric masking pipeline can solve these visual and hardware constraints on standard consumer-grade GPUs.

3. Related Work: Foundational Architectures

- **3.1 Neural Radiance Fields (NeRF)**
- **3.2 DietNeRF**
- **3.3 RegNeRF**

I will analyze foundational continuous volume rendering architectures and their evolution into few-shot conditional models. The discussion will focus specifically on how these earlier frameworks struggled with evaluation bias and computational overhead during live-fire spatial deployments.

3. Related Work: Rapid Generative Protocols

- **3.4 DreamGaussian**
- **3.5 Large Gaussian Models (LGM)**
- **3.6 The Trellis Protocol**

I will then explore cutting-edge generative frameworks that successfully weaponized 3D Gaussians for pure generative tasks. I will detail how these protocols prioritize blistering inference speeds and compact structured latents to achieve rapid text-to-3D and image-to-3D conversions.

4. Methodology

- **Asynchronous Architecture & Deep Splat Excavation**

I will detail the strictly decoupled client-server architecture built to isolate the lightweight presentation viewport from heavy backend tensor operations. I will also explain the logic of the Deep Splat Excavation (DBSE) algorithm, which non-destructively hole-punches the Gaussian environment to inject assets without visual collision.

5. Discussion and Limitations

- **VRAM Constraints and Generative Fragility**

I will transparently address the strict 8GB VRAM limits of the edge hardware and the pipeline latency introduced by sequential memory flushing. I will also evaluate the system's sensitivity to semantic masking noise, noting how imperfect background isolation leads to flattened, corrupted geometric outputs.

6. Conclusion

- **Establishing Secure Spatial Computing**

I will summarize how the Vectra Protocol successfully bridged edge-computed AI with interactive digital twins without relying on cloud computation. Finally, I will outline how this methodology establishes the necessary groundwork for embedding rigorous mathematical safety mechanisms into future spatial generations.

7. References

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- Niemeyer, M., et al. "RegNeRF: Regularizing Neural Radiance Fields for View Synthesis from Sparse Inputs." *CVPR*, 2022.
- Tang, J., et al. "DreamGaussian: Generative Gaussian Splatting for Efficient 3D Content Creation." *ICLR*, 2024.
- Tang, J., et al. "LGM: Large Multi-View Gaussian Model for High-Resolution 3D Content Creation." *arXiv preprint*, 2024.
- Xiang, B., et al. "Trellis: Native and Compact Structured Latents for 3D Generation." *arXiv preprint*, 2025.